

QUINULT 4 NE, WA, USA

WMO: 742030

Lat: 47.5139N

Lon: 123.8120W

Elev: 87

StdP: 100.28

Time Zone: -8.00 (NAP)

Period: 06-19

WBAN: 04237

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.4	-2.2	-6.8	2.1	1.8	-4.0	2.7	-0.2	2.8	4.3	2.2	4.8	0.3	N/A	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	28.6	18.8	26.1	18.1	23.9	17.2	20.0	27.3	19.0	25.4	18.0	23.4	1.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.5	22.2	16.5	11.9	21.0	15.8	11.3	19.9	57.1	27.1	54.0	25.4	51.1	23.4	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
2.1	1.9	1.4	-5.6	33.8	1.9	1.9	-7.0	35.2	-8.1	36.4	-9.1	37.5	-10.4	38.9		
			-5.1	22.1	1.8	0.6	-6.4	22.5	-7.4	22.8	-8.4	23.2	-9.7	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	4.1	5.1	6.2	8.2	11.7	14.0	16.8	17.3	14.8	10.4	6.3	3.6
	DBStd	5.57	3.01	2.85	2.83	3.11	3.18	2.94	2.43	2.60	2.77	2.39	3.19	3.01
	HDD10.0	870	184	137	122	71	17	1	0	0	1	23	115	199
	HDD18.3	3121	443	369	376	304	208	134	62	52	111	246	359	457
	CDD10.0	837	1	1	4	17	69	122	209	227	145	36	6	1
	CDD18.3	45	0	0	0	1	1	6	13	19	5	0	0	0
	CDH23.3	603	0	0	2	11	42	79	187	216	67	1	0	0
	CDH26.7	177	0	0	0	3	7	23	58	69	17	0	0	0
Wind	WSAvg	0.5	0.6	0.6	0.6	0.6	0.6	0.5	0.5	0.5	0.4	0.5	0.6	0.6
Precipitation	PrecAvg	3572	537	405	402	245	155	101	79	95	154	333	473	593
	PrecMax	4434	820	735	759	428	283	210	240	282	371	840	696	912
	PrecMin	2963	246	202	86	91	50	21	14	4	2	34	175	284
	PrecStd	522	152	164	168	104	67	58	70	76	100	204	147	159
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	15.0	20.5	24.4	27.6	30.3	32.1	32.4	29.2	21.6	14.4	11.8
		MCWB	10.9	10.7	11.8	14.4	16.1	20.8	19.6	19.9	18.8	14.4	13.6	11.1
	2%	DB	10.3	12.3	15.2	19.3	23.8	25.3	27.7	28.4	25.0	17.8	13.0	10.0
		MCWB	9.8	10.1	9.6	12.6	15.6	18.1	19.0	19.3	17.5	14.0	12.4	9.5
	5%	DB	9.3	10.6	12.2	15.6	20.3	22.3	25.0	25.5	22.1	15.6	12.2	8.9
		MCWB	8.8	9.5	8.9	10.7	14.8	16.6	18.2	18.4	16.7	13.3	11.7	8.5
	10%	DB	8.3	9.3	10.4	13.0	17.6	19.9	22.7	22.9	19.5	14.1	11.0	7.8
		MCWB	7.9	8.6	8.7	9.6	13.2	15.3	17.0	17.3	15.7	12.7	10.5	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	12.5	12.7	16.0	17.7	21.2	20.4	21.0	20.6	17.2	14.0	11.7
		MCDB	11.7	13.3	17.4	24.0	25.1	28.9	28.0	30.0	28.1	20.1	14.2	12.0
	2%	WB	10.2	10.8	11.0	13.7	16.3	19.3	19.4	19.9	18.6	15.1	12.7	9.8
		MCDB	10.5	11.6	14.5	19.1	22.7	25.7	26.6	27.2	23.7	16.5	12.9	10.1
	5%	WB	9.1	9.9	10.1	11.7	15.1	17.6	18.5	18.9	17.5	13.9	11.8	8.7
		MCDB	9.3	10.6	12.6	15.3	21.0	22.3	25.1	25.3	21.5	14.9	12.2	8.9
	10%	WB	8.0	8.9	9.2	10.5	13.9	16.0	17.4	17.9	16.4	12.9	10.6	7.4
		MCDB	8.3	9.7	10.9	12.8	17.9	19.9	22.7	23.1	19.1	13.9	11.1	7.7
Mean Daily Temperature Range	5% DB	MDBR	4.8	6.1	7.4	8.6	10.1	10.2	11.7	11.5	9.9	7.7	5.4	4.4
		MCDBR	5.4	8.5	12.0	14.2	15.8	15.2	16.3	16.5	15.0	10.9	6.3	4.8
		MCWBR	5.2	5.5	6.9	7.7	8.3	8.1	8.1	7.9	7.9	6.9	5.3	4.3
	5% WB	MCDBR	5.4	6.7	9.7	12.7	14.7	14.6	15.2	15.3	12.8	8.6	5.5	4.6
		MCWBR	5.0	5.2	6.0	7.2	8.0	8.1	7.8	7.6	7.1	6.2	5.0	4.4
Clear-Sky Solar Irradiance	taub	0.326	0.324	0.338	0.371	0.403	0.376	0.360	0.362	0.348	0.339	0.335	0.324	
	taud	2.463	2.505	2.482	2.360	2.277	2.391	2.465	2.460	2.500	2.518	2.443	2.423	
	Ebn at Noon	755	846	887	882	861	887	897	882	864	815	732	703	
	Edh at Noon	66	78	92	114	128	115	106	102	89	76	67	62	
All-Sky Solar Radiation	RadAvg	1.00	1.88	2.85	4.18	5.21	5.69	6.15	5.26	3.93	2.24	1.19	0.83	
	RadStd	0.14	0.24	0.32	0.38	0.42	0.55	0.47	0.37	0.35	0.24	0.15	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page